Image Quality Metrics for Motion Assessment

ISMRM Workshop on

Motion Correction in MR

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KØBENHAVNS UNIVERSITET

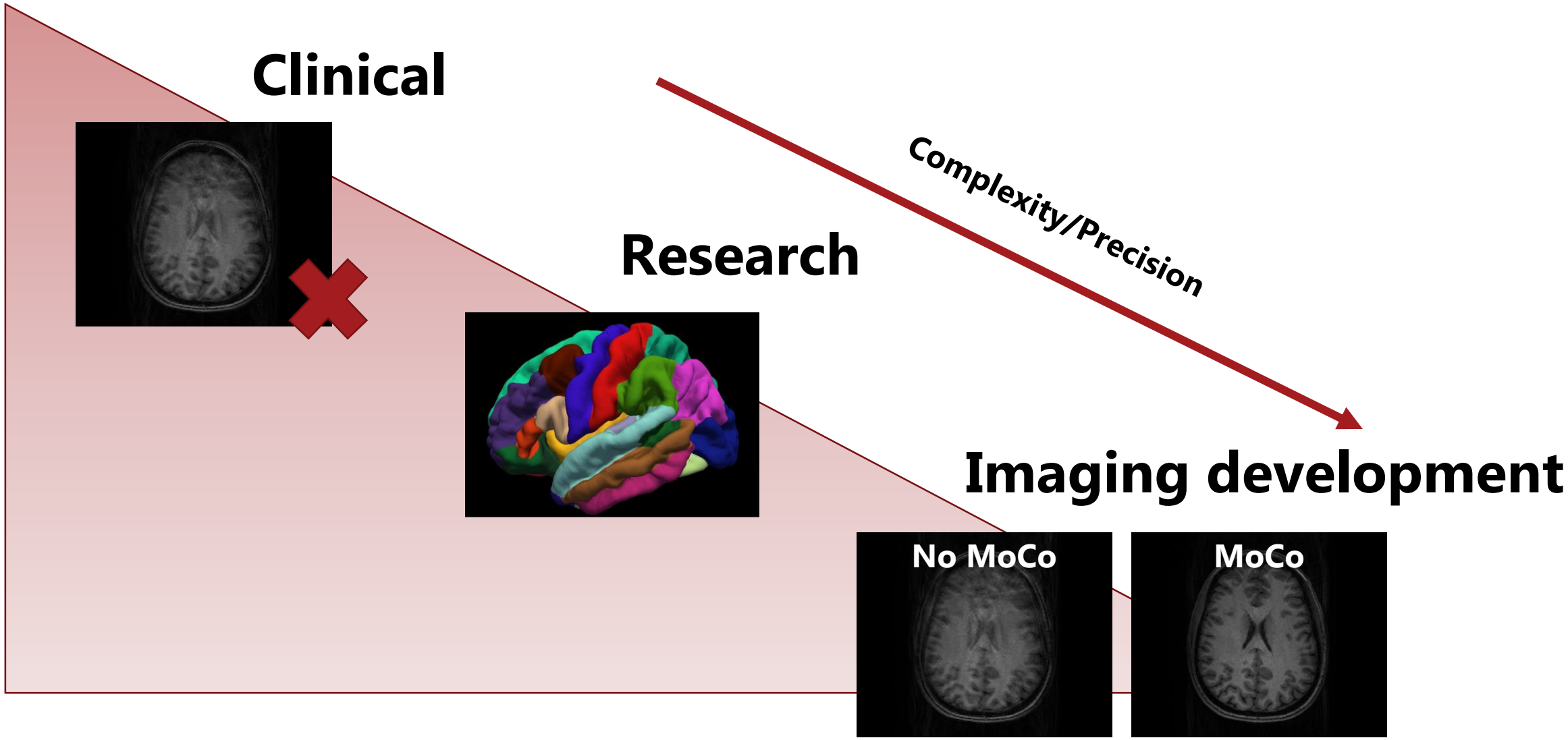
Declaration of Financial Interests or Relationships

Melanie Ganz-Benjaminsen:

I have no financial interests or relationships to disclose with regard to the subject matter of this presentation.

AL
5cm
MF 1.08
TR 20.0
TE 8.9
TA 03:25
BW 60.0
M/ND
A1
HL1,2;HS1,2
*fl3d1 / 30
SP L7.8
SL 5.0
FoV 230*230
192*256
Sag>Cor(-3.2)>Tra(1.6)
W 854
C 407
SP L12.8
W 880
C 423
17:10:54
2 IMA 17 / 10
AF
HOSPITAL
MR 2004C
HLS
+LPH
1043
*27/09/1975, F, 33Y
STUDY 1
26/03/2009
17:19:59
4 IMA 5 / 9
4 IMA 7 / 9
17:20:02
4 IMA 8 / 9
R
MF 1.14
TR 4720.0
TE 108.0
TA 03:27
BW 70.0
M/ND
SP F17.7
SP F4.7
FoV 194*230
190*256s

Applications of Image Quality Metrics (IQMs)

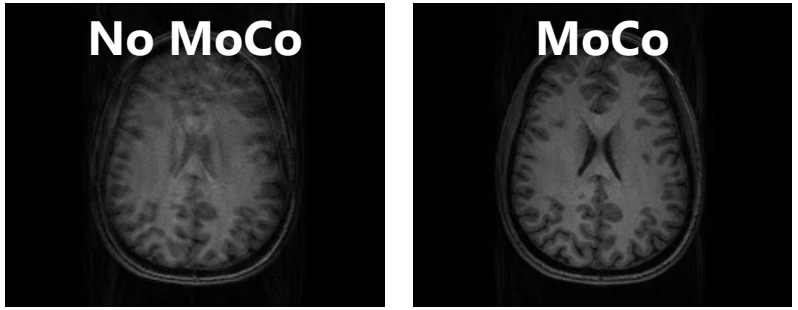


Types of IQMs

- Qualitative vs. quantitative

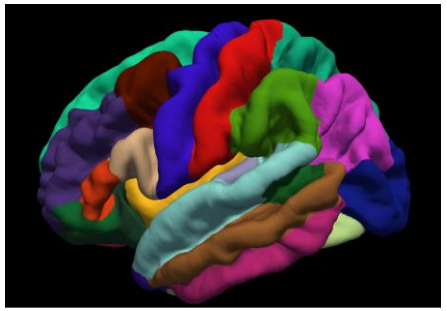
Average Edge Strength = 0.6

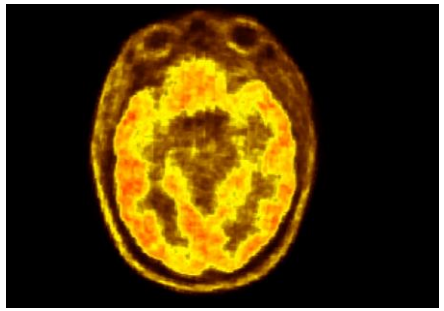
- Reference-based vs. reference-free



Patient on table

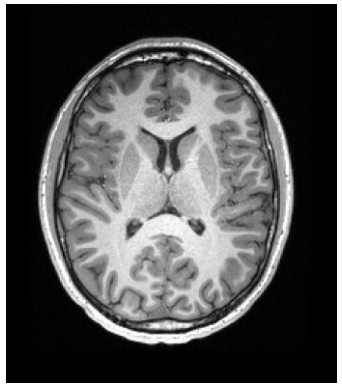
- Task-based





[van der Kouwe & Jalal 2022, Chapter 8]

How do we get a quantitative IQM?



	Metric	Definition
Full-reference	SSIM	$\frac{1}{\lvert\mathcal{M}\rvert}\sum_{m,\hat{m}\in\mathcal{M}}\frac{(2\mu_m\mu_{\hat{m}}+c_1)(2\sigma_{m\hat{m}}+c_2)}{(\mu_m^2+\mu_{\hat{m}}^2+c_1)(\sigma_m^2+\sigma_{\hat{m}}^2+c_2)}$
	PSNR	$10\log_{10}\frac{\max(\hat{x})^2}{\frac{1}{IJ}\sum_{i=1,j=1}^{I,J}(x_{ij}-\hat{x}_{ij})^2}$
	FSIM	*Due to the complexity, please refer to Zhang et al.*[19]
	VIF	*Due to the complexity, please refer to Sheikh et al.*[20].
	LPIPS	$\mathbf{d}(\mathcal{F}(x),\mathcal{F}(\hat{x}))$
Reference-free	TG	$\frac{1}{IJ}\sum_{i=1,j=1}^{I,J}g_{i,j}^2$
	IE	$-\sum_{i,j}y_{i,j}\ln(y_{i,j})$ `with` $y_{i,j}=\frac{x_{i,j}}{\sqrt{\sum x_{i,j}^2}}$
	AES	$\frac{\sqrt{\sum_{i,j}E(x_{i,j})g_{i,j}^2}}{\sum_{i,j}E(x_{i,j})}$
	NGS	$(\frac{g_{i,j}}{\sum_{i,j}g_{i,j}})^2$
	GE	$-\sum_{i,j}z_{i,j}\ln(z_{i,j})$ `with` $z_{i,j}=\frac{g_{i,j}}{\sqrt{\sum g_{i,j}^2}}$

Simple, right?

Article | Published: 20 May 2020

Variability in the analysis of a single neuroimaging dataset by many teams

Rotem Botvinik-Nezer, Felix Holzme

Nature **582**, 84–88 (2020) | Cite thi

44k Accesses | **232** Citations | **20**

Review Article

JCBFM

Journal of Cerebral Blood Flow & Metabolism
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DOI: 10.1177/0271678X18770107
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SAGE

Cerebral serotonin transporter measurements with [^{11}C]DASB: A review on acquisition and preprocessing across 21 PET centres

Martin Nørgaard[1,2], Melanie Ganz[1,3], Claus Svarer[1], Masanori Ichise[4], Rupert Lanzenberger[5], Mark Lub Ramin V Parsey[7], Marios Politis[8], Eugenii A Rabine Mark Slifstein[7], Vesna Sossi[11], Tetsuya Suhara[4], Peter S Talbot[12], Federico Turkheimer[13], Stephen C and Gitte M Knudsen[1,2]

HUMAN BRAIN MAPPING

Influence Series

Research Article | Full Access

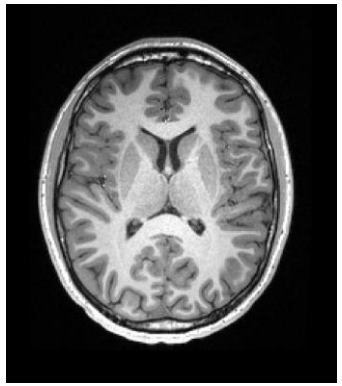
Optimizing preprocessing and analysis pipelines for single-subject fMRI. I. Standard temporal motion and physiological noise correction methods

Nathan W. Churchill, Anita Oder, Hervé Abdi, Fred Tam, Wayne Lee, Christopher Thomas, Jon E. Ween, Simon J. Graham, Stephen C. Strother

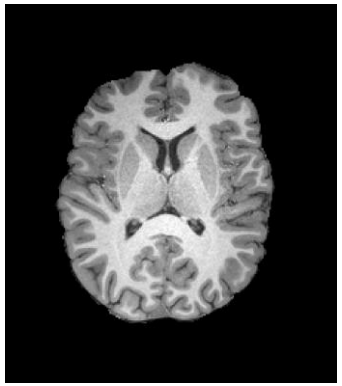
First published: 31 March 2011 | https://doi.org/10.1002/hbm.21238 | Citations: 76

Brainmask

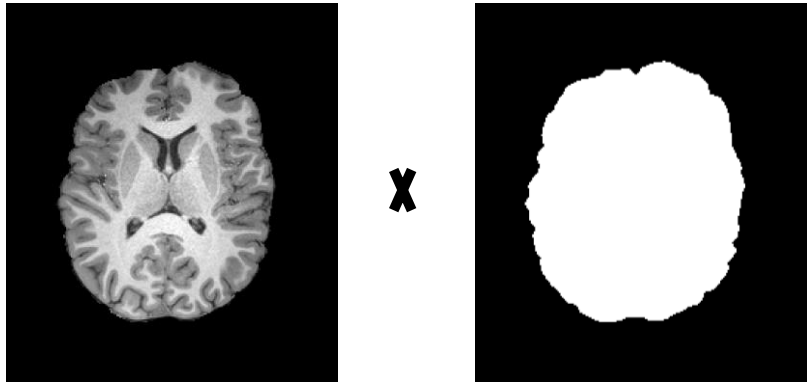
- None



- Mask

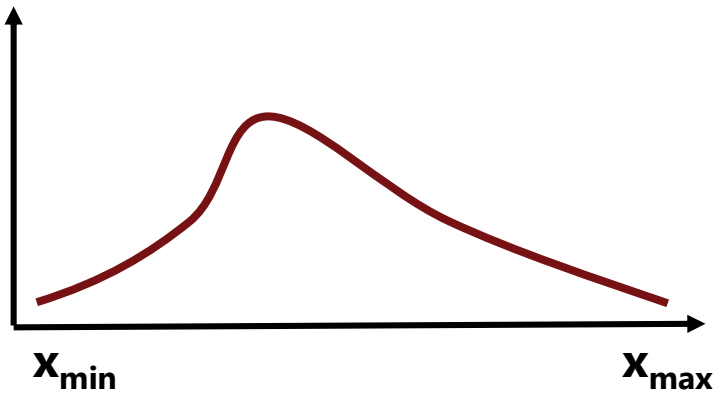


- Multiply

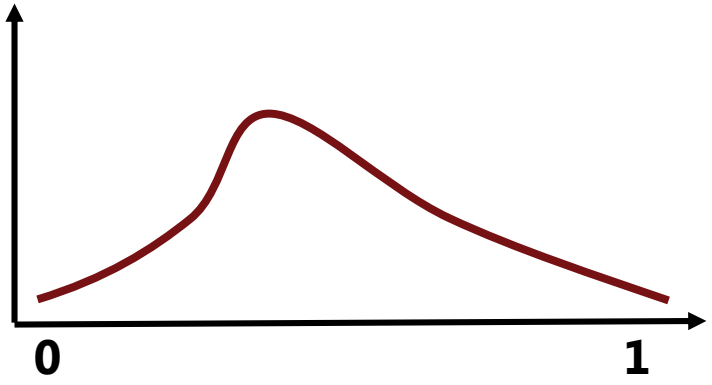


Normalization

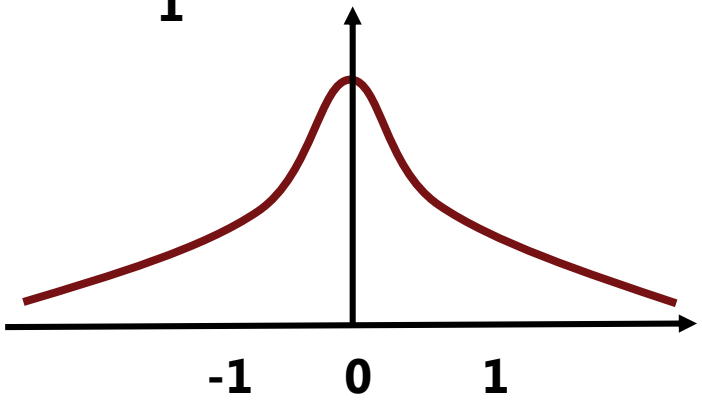
- None



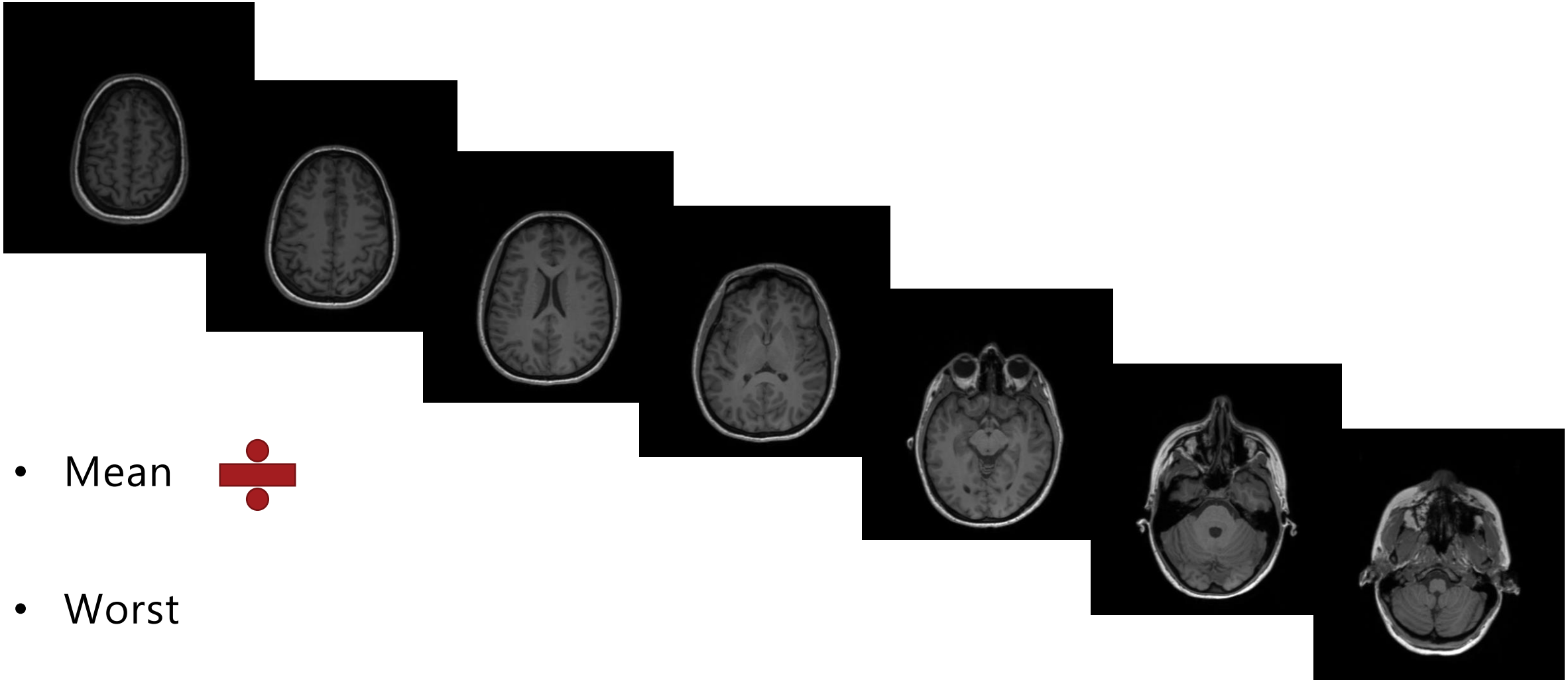
- Min-max



- Mean-std

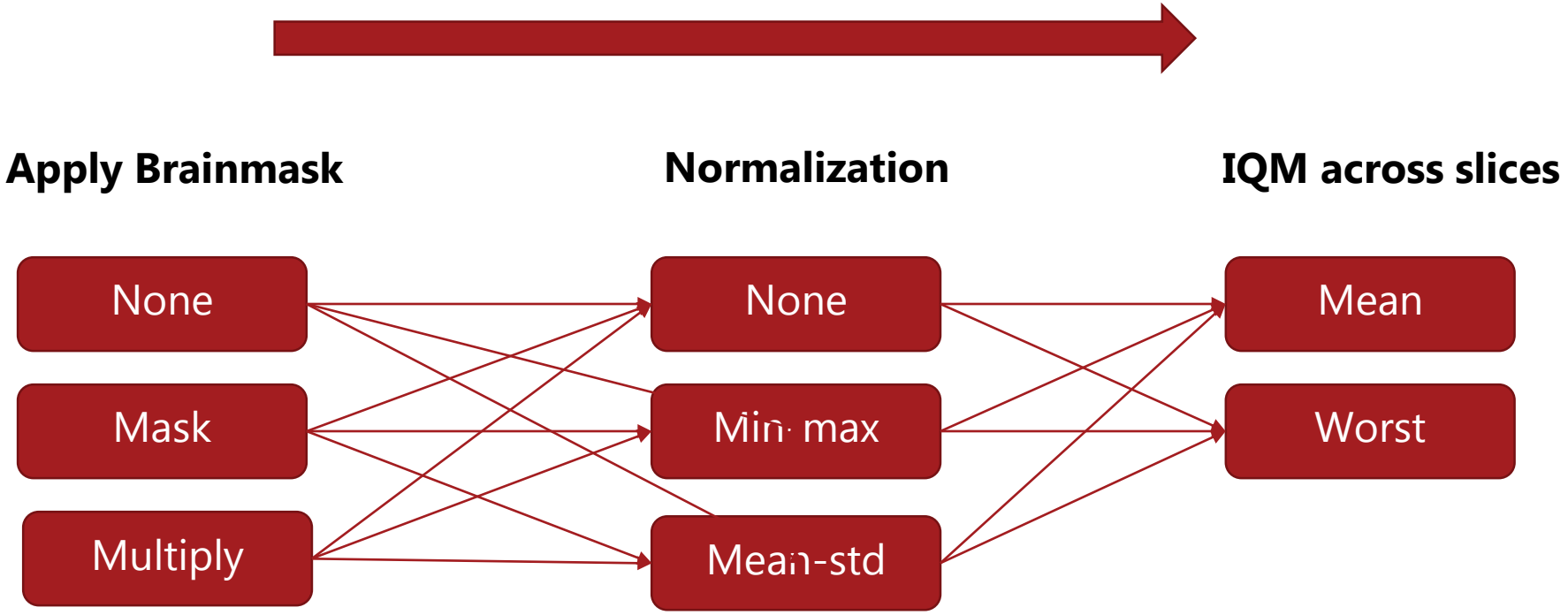


IQM across slices



- Mean ÷
- Worst

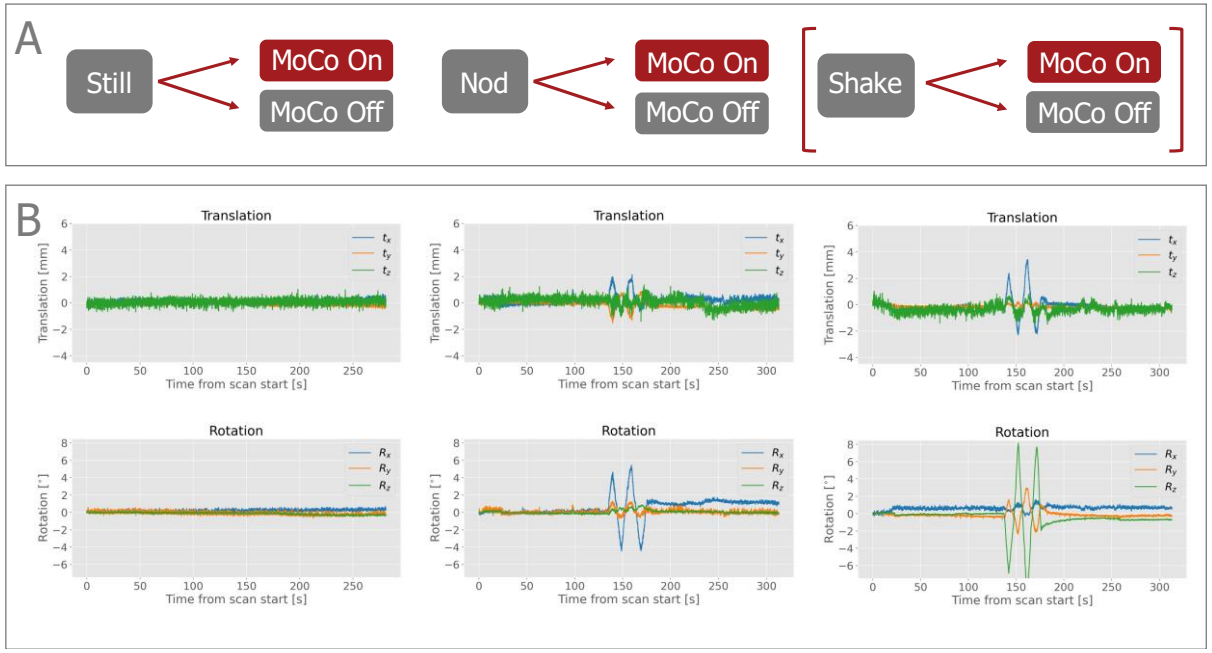
Choices in preprocessing



This is a non-exhaustive list! -> registration, brainmask derivation, etc.

Data - MR study about motion correction

- 22 healthy volunteers
- Four acquisitions for each sequence of a clinical protocol (six for MPRAGE)
- Motion degraded as well as motion free data



https://openneuro.org/datasets/ds004332/

MR Sequences

	T_1 MPR	T_2 FLAIR	T_1 STIR	T_2 TSE
T_R [ms]	2000	5000	4100	4010
T_I [ms]	900	1800	1600	—
T_E [ms]	2.32	388	8.5	116
α [°]	8.0	variable	150.0	150.0
Voxel size [mm]	0.9×0.9×0.9	1.0×1.0×1.0	0.9×0.9×5.0	0.4×0.4×5.0
Orientation[1]	sagittal	sagittal	transversal	transversal
PE Direction	A → P	A → P	R → L	R → L
Scan duration [s]	280	252	190	150

3D-encoded 2D-encoded

Quantitative and qualitative IQMs

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Full-reference	SSIM	$\frac{1}{\lvert\mathcal{M}\rvert}\sum_{m,\hat{m}\in\mathcal{M}}\frac{(2\mu_m\mu_{\hat{m}}+c_1)(2\sigma_{m\hat{m}}+c_2)}{(\mu_m^2+\mu_{\hat{m}}^2+c_1)(\sigma_m^2+\sigma_{\hat{m}}^2+c_2)}$
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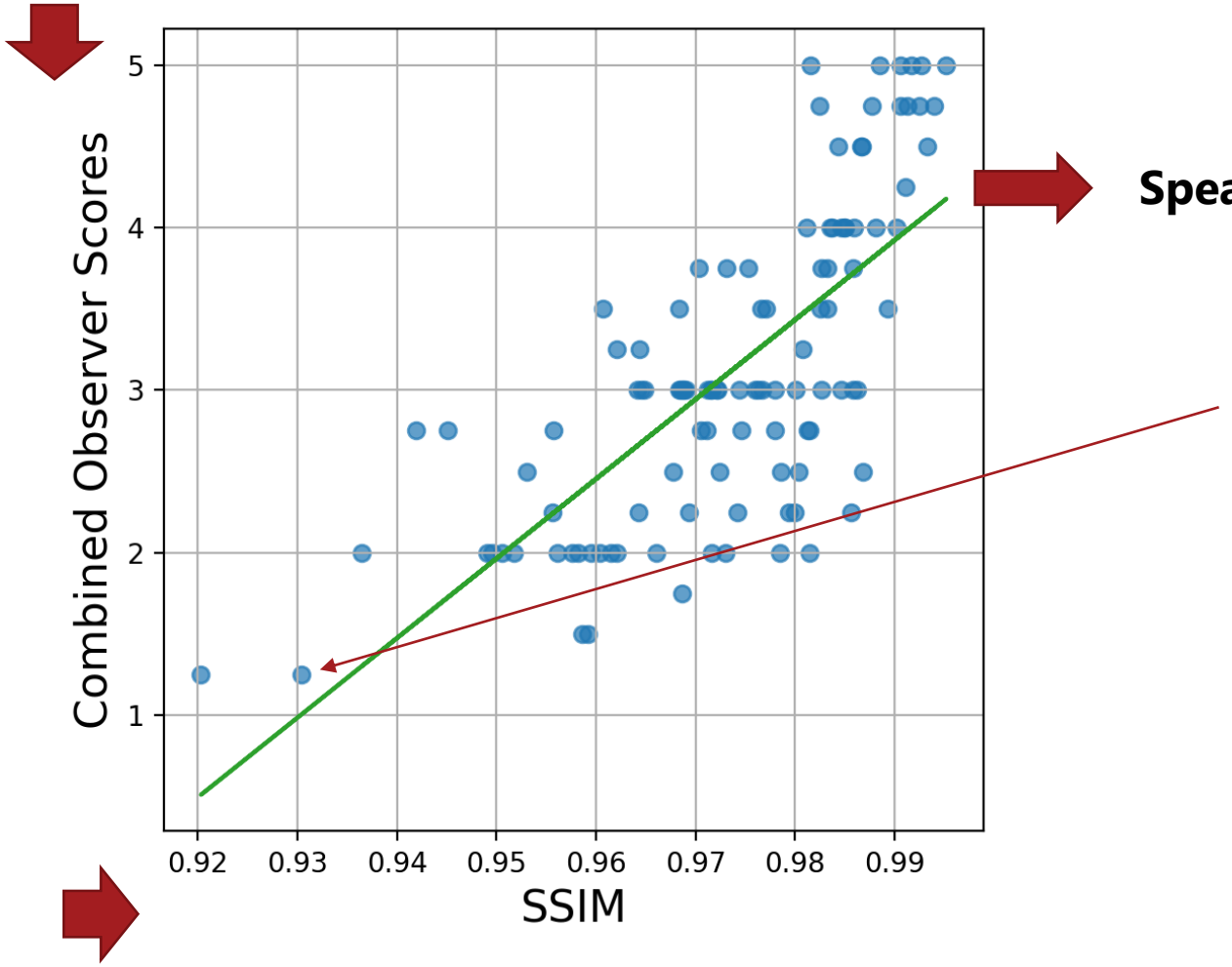
	GE	$-\sum_{i,j} z_{i,j}\ln(z_{i,j})$ `with` $z_{i,j}=\frac{g_{i,j}}{\sqrt{\sum g_{i,j}^2}}$

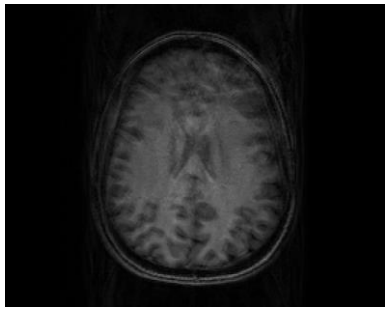
[Spieker et al. IEEE TMI 2024]

Likert-scale evaluation by 4 observers

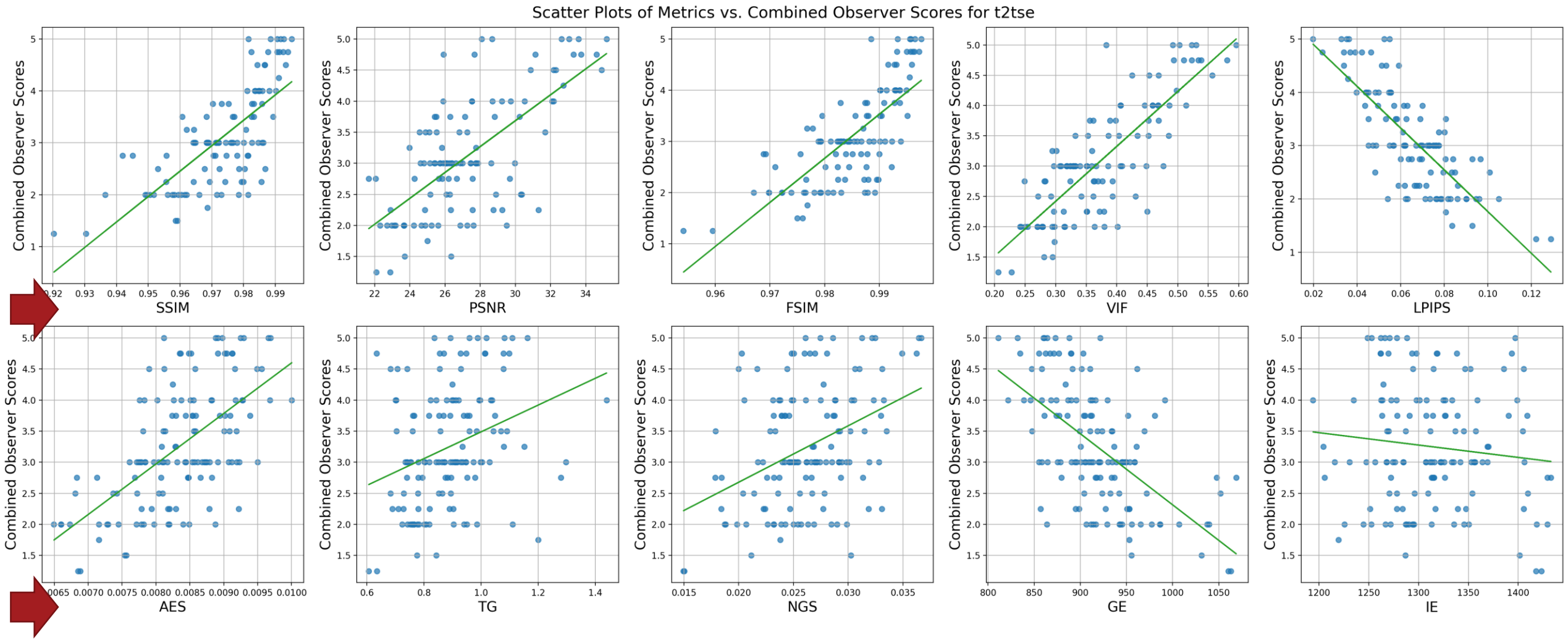
Relation to observer scores



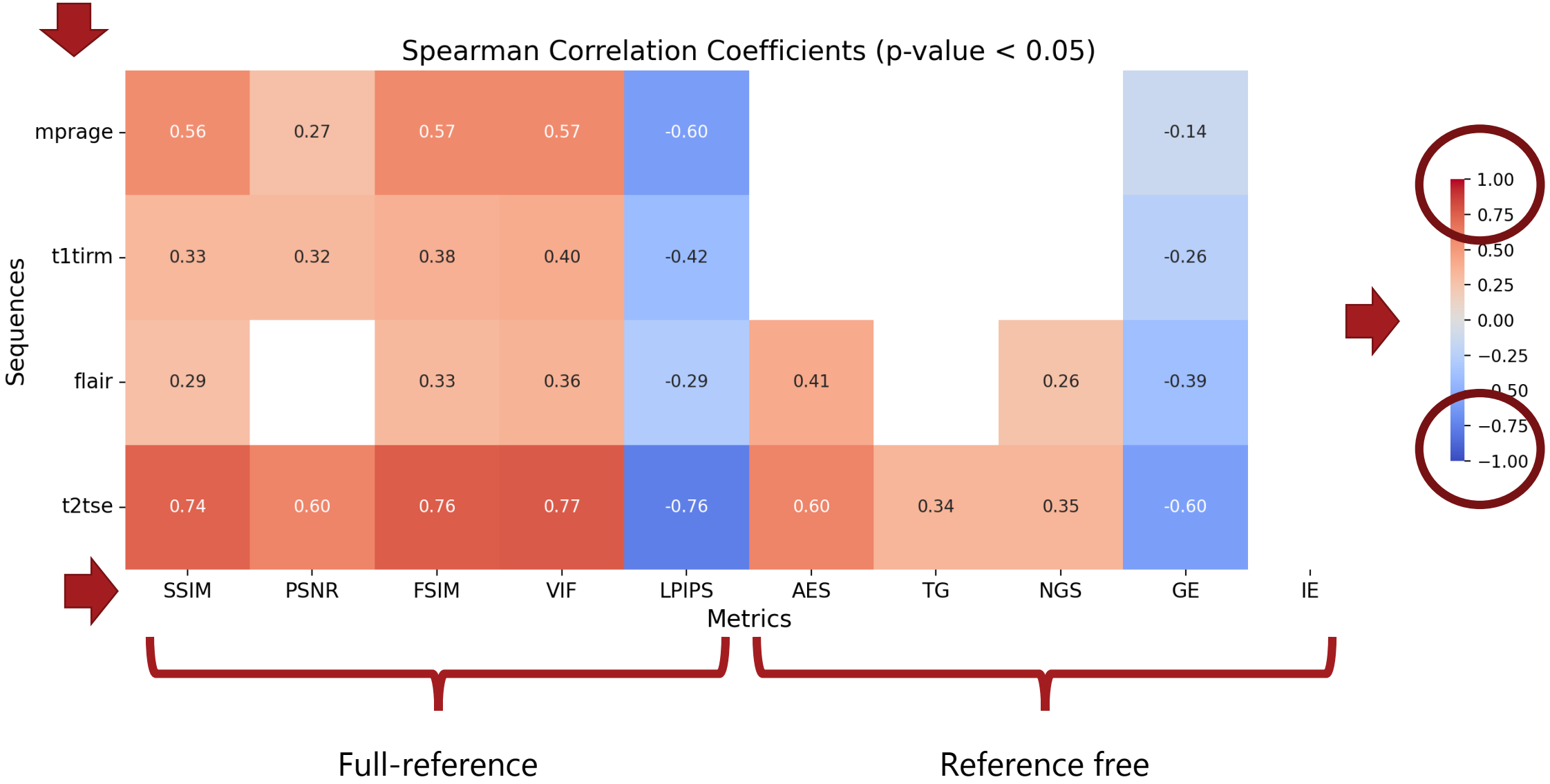
Spearman rank correlation



Relation to observer scores for all quantitative IQMs

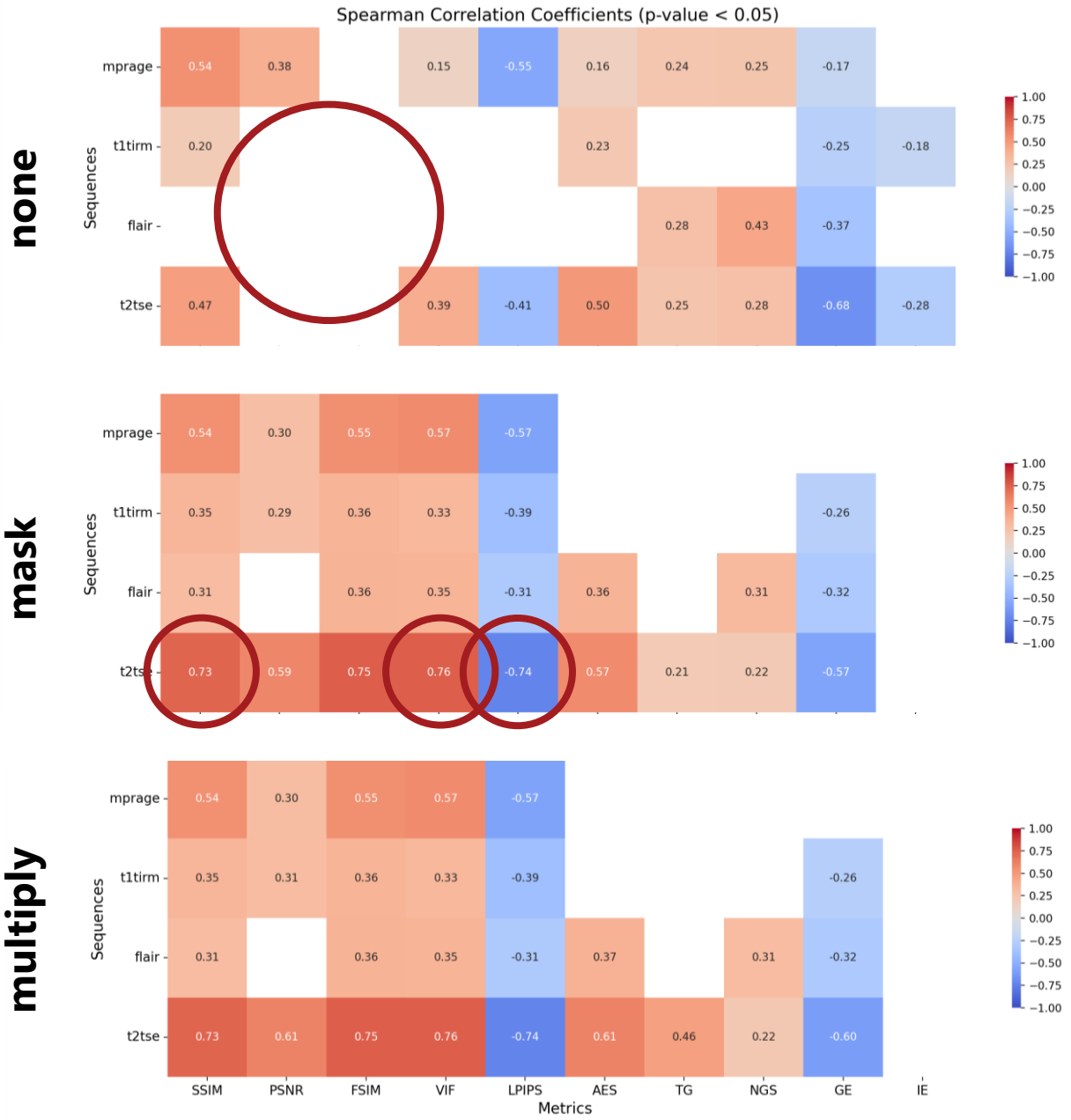


How to evaluate IQM values



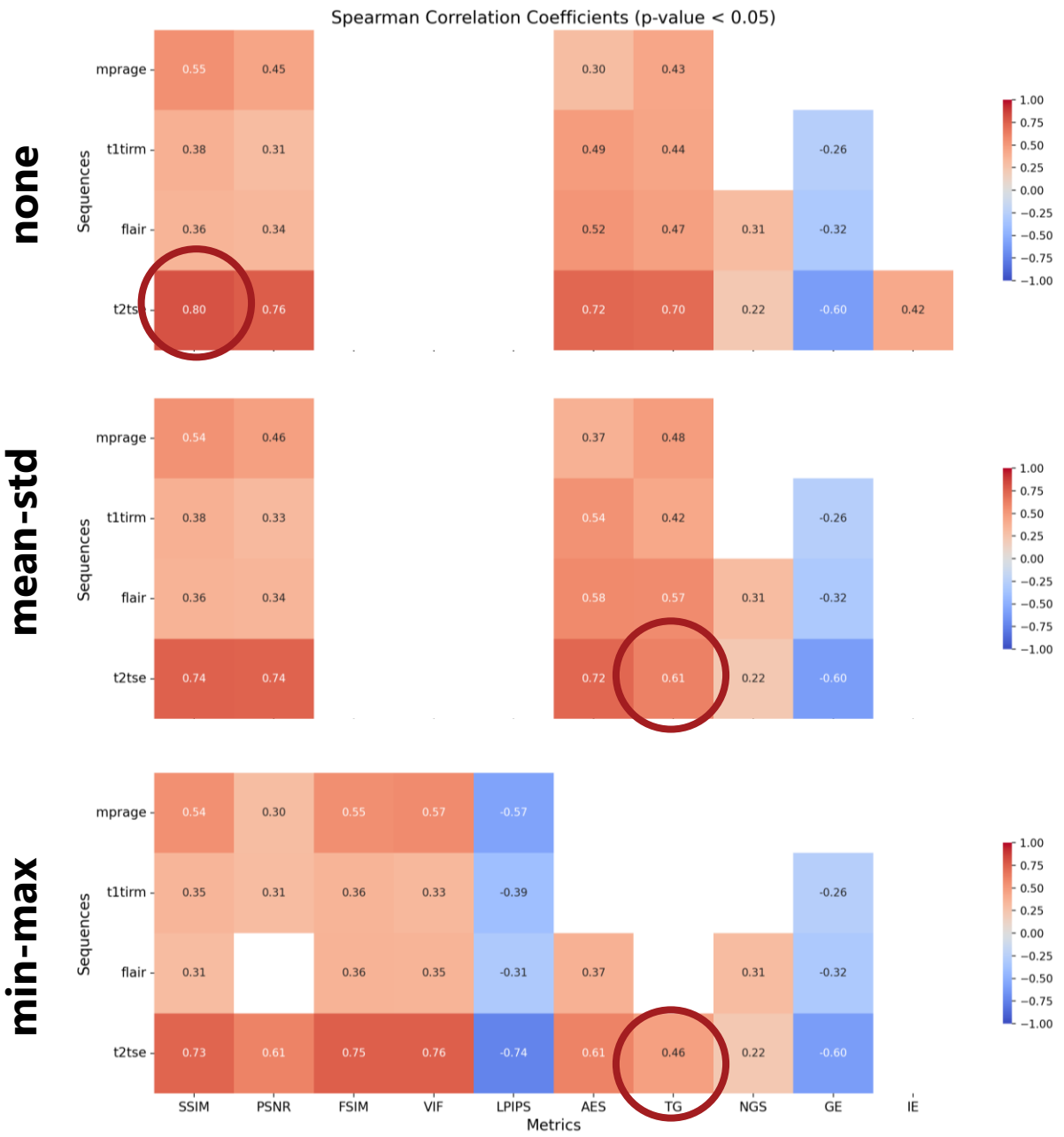
Influence of the brainmask

- How high do our correlations need to be?
- No brainmask -> less power
- Other settings:
  - min-max normalization
  - worst slice



Effect of normalization

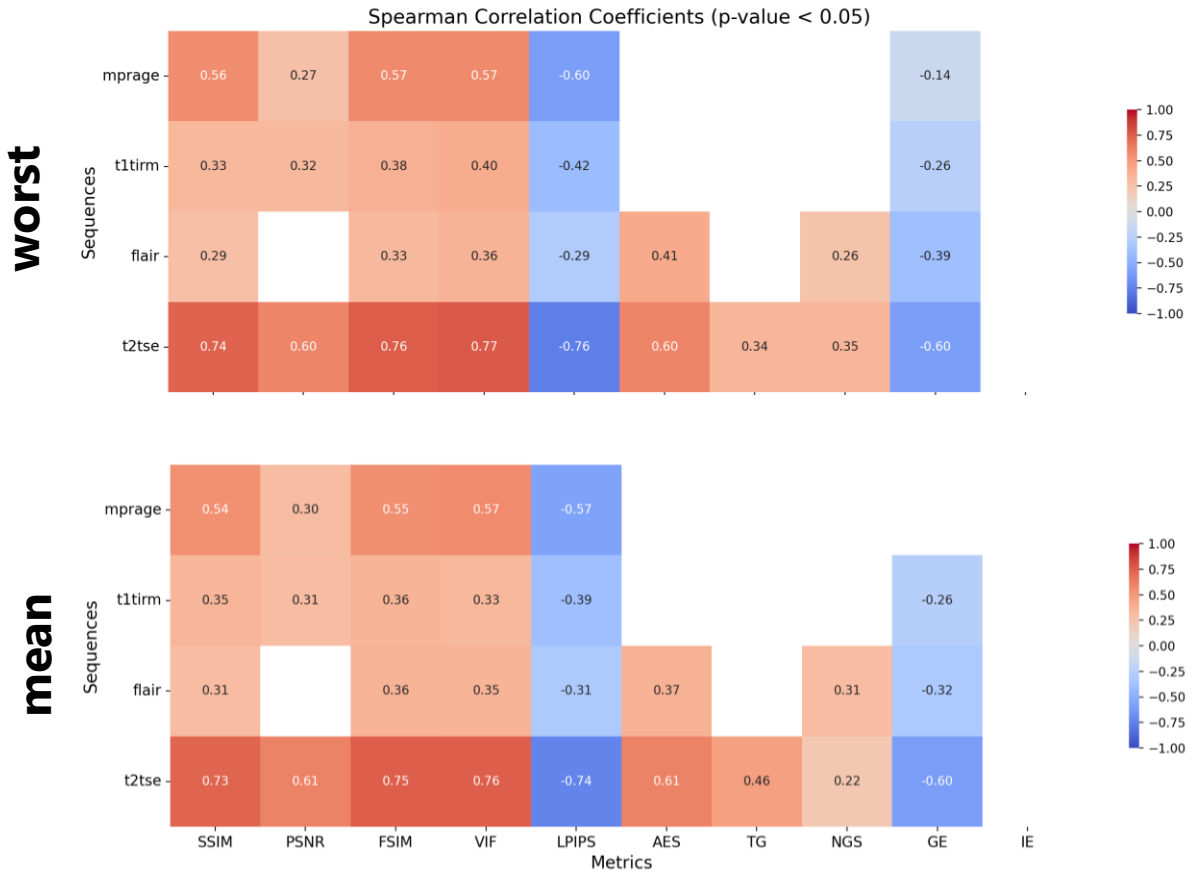
- FSIM, VIF and LPIPS require min-max
- Large difference for some non-reference based methods
- Other settings:
  - brainmask multiply
  - worst slice



Effect of summarizing the final IQM

- No large effect of summarization -> also due to scoring instructions of observers

- Other settings:
  - brainmask multiply
  - min-max normalization



Discussion & Limitations

- Comparing metric values between publications is impossible!
- Need for standards of reporting & best practices
- Only evaluated for rigid motion aka brain
- Data used is open data with real motion degradation – not simulated!

Image Quality Team & Funding

Neurobiology Research Unit
Rigshospitalet
Copenhagen University Hospital

REGION H

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elsass foundation

MUDS Munich School for Data Science HELMHOLTZ · TUM · LMU

Center for Biomedical Imaging

NYU Langone Health

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Come talk to me!

Code will be released on GitHub!